1. Difference Between OLTP and OLAPOLTP (Online Transaction Processing) and OLAP (Online Analytical Processing) are two fundamental types of database processing systems designed for entirely different workloads.OLTP manages day-to-day, operational business transactions as they happen (e.g., a customer buying an item on an e-commerce site or withdrawing cash from an ATM).OLAP handles complex queries, data aggregation, and business intelligence reporting to help management make strategic decisions (e.g., analyzing regional sales trends over the past five years).Comparison TableFeatureOLTP (Online Transaction Processing)OLAP (Online Analytical Processing)Primary PurposeTo run and manage daily business operations and transactions.To support data analysis, business intelligence, and decision-making.Data ContentCurrent, up-to-date, detailed transactional data.Historical, aggregated, multi-dimensional data.Database DesignHighly normalized (3NF) to reduce data redundancy and ensure fast writes.Denormalized (using Star or Snowflake schemas) to optimize read performance.Query TypeSimple, short reads and writes (CRUD operations: Create, Read, Update, Delete).Complex, read-heavy queries involving multi-table joins and aggregations.UsersClerks, cashiers, customers, and front-line operational staff.Business analysts, data scientists, managers, and executives.Response TimeMilliseconds (fast execution for individual records).Seconds to minutes (due to heavy data scanning and processing).Data VolumeSmall to moderate data processed per transaction.Massive volumes of historical data queried at once.2. Difference Between OLTP and Data

WarehousingTo understand this comparison, it helps to know that OLTP is a processing system, whereas a Data Warehouse is a storage and repository system. They are often connected: data from an OLTP system is periodically extracted, transformed, and loaded (ETL) into a Data Warehouse to be analyzed using OLAP tools.OLTP captures live transactions in real-time.Data Warehousing (DWH) centralizes historical data from multiple sources into a single, unified repository optimized for reporting.Comparison TableFeatureOLTP (Online Transaction Processing)Data Warehousing (DWH)Core NatureA processing system (application-oriented).A storage repository (subject-oriented).Data SourcesUser input, web forms, POS terminals, and operational applications.Multiple OLTP databases, flat files, APIs, and external sources.State of DataVolatile and mutable (frequently updated, deleted, or overwritten).Non-volatile (read-only, historical records that are appended rather than changed).Performance FocusMaximizing transaction throughput (handling thousands of concurrent writes/reads).Maximizing query performance and reporting speed over large datasets.Storage SizeRelatively smaller (often stores recent operational data, e.g., last 30–90 days).Extremely large (stores years or decades of historical enterprise data).